

Numerical methods, ex.2 ch.19.2 p.798

Find a solution for $f(x) = x^3 + x - 1 = 0$ by iteration.

Solution:

$$(I) \quad x^3 + x - 1 = 0 \Rightarrow x = \underbrace{1 - x^3}_{g_1(x)}$$

$$(II) \quad x^3 + x - 1 = 0 \Rightarrow x(x^2 + 1) = 1$$

$$\begin{aligned} \frac{d}{dx} (x^2 + 1)^{-1} &= -1 \cdot 2x(x^2 + 1)^{-2} \\ &= \frac{-2x}{(x^2 + 1)^2} \end{aligned} \Rightarrow x = \underbrace{\frac{1}{x^2 + 1}}_{g_2(x)}$$

$$(I) \quad g_1'(x) = -3x^2 \Rightarrow |g_1'(x)| > 1$$

Don't meet conditions!

$$(II) \quad g_2'(x) = \frac{-2x}{(x^2 + 1)^2} \Rightarrow |g_2'(x)| = \frac{2|x|}{(x^2 + 1)^2}$$

Here we have $|g_2'(x)| \leq K < 1$
3rd condition met.

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1st condition also met; the function is differentiable.

2nd condition; does it have a solution?

$f(0) = -1$, $f(1) = 1$; the function changes sign on the interval $I = \{0, 1\}$, it is continuous, all of which indicates there is a solution.

Conditions are met, let's iterate; selecting $x_0 = \frac{1}{2}$

$$x_0 = \frac{1}{2}$$

$$x_1 = g_2(x_0) = 0.8$$

$$x_2 = g_2(x_1) = 0.6097560976 \left(\frac{25}{41}\right)$$

$$x_3 = g_2(x_2) = 0.7289679098 \left(\frac{1681}{2306}\right)$$

$$x_4 = g_2(x_3) = 0.6529997248$$

$$x_5 = g_2(x_4) = 0.701061373$$

$$x_6 = g_2(x_5) = 0.6704717958$$

$\vdots$

$$x_{20} = g_2(x_{19}) = 0.6823070721$$

True solution ~ 0.682327803828

$$\begin{aligned} \text{Error} &= 0.682327803828 - 0.6823070721 = 0.000020731728 \\ &= 0.20731728 \cdot 10^{-5} \end{aligned}$$

$$h(x) = |g'_2(x)| = \frac{2|x|}{(x^2+1)^2}$$

$\mathcal{I} = [0, 1]$, so we can simplify

$$h(x) = \frac{2x}{(x^2+1)^2}$$

Now we take the derivative of $h(x)$ to find its maximum, which also is our K (kappa).

Applying quotient rule:

$$u = 2x$$

$$u' = 2$$

$$v = (x^2+1)^2$$

$$v' = 4x(x^2+1)$$

$$u'v = 2 \cdot 2(x^2+1)^2 = 2x^4 + 4x^2 + 2$$

$$uv' = 2x(4x(x^2+1)) = 8x^2(x^2+1)$$

$$u'v - uv' = 2(x^2+1)^2 - 8x^2(x^2+1)$$

$$= 2(x^2+1)[(x^2+1) - 4x^2]$$

$$v^2 = (x^2+1)(x^2+1)^3$$

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

$$h'(x) = \frac{u'v - uv'}{v^2} = \frac{2(x^2+1)[(x^2+1) - 4x^2]}{(x^2+1)^4}$$

$$= \frac{2((x^2+1) - 4x^2)}{(x^2+1)^3} = \frac{2x^2 + 2 - 8x^2}{(x^2+1)^3}$$

$$h'(x) = \frac{2(1-3x^2)}{(x^2+1)^3}$$

Finding the maximum: $h'(x) = 0$ when the numerator equals 0.

$$2 - 6x^2 = 0 \Rightarrow 6x^2 = 2 \Rightarrow x^2 = \frac{1}{3}$$

$$x = \pm\sqrt{\frac{1}{3}}$$

Since we're dealing with absolutes, $x = \frac{1}{\sqrt{3}}$

$$\text{Finding } K: K = h\left(\frac{1}{\sqrt{3}}\right) = \frac{3\sqrt{3}}{8} \approx 0.649519$$

$$\text{In } \mathcal{I} \quad K = \max_{x \in [0, 1]} |g'_2(x)| = h(x_{\text{critical}}) = \frac{3\sqrt{3}}{8} \sim 0.64... < 1$$

We should also check the endpoints:

$$h(0) = 0$$

$$h(1) = \frac{1}{2}$$

This gives: $h(x_{\text{critical}}) > h(1) > h(0)$